Modeling the stock market using the chaos theorem

Ishaan Gowda

The International School Bangalore

Aim of the Exploration

This exploration aims to determine whether the chaos theory can predict the stock market. I seek to comprehensively understand how to model and predict the stock market accurately.

Plan of Investigation

I will use quadratic mapping and calculus to model the stock market for this investigation. I will select two stocks from the Yahoo Finance Website to analyze. I am a technology enthusiast. Hence, I will examine two stocks in the technology sector: Microsoft Corporation (MSFT) and Amazon.com Inc (AMZN). Since the period between 2019 and 2021 was affected by COVID-19 and may fail to provide a mirage of the stock performance, I decided to collect the latest data between 2022 and 2023. I will evaluate the model's accuracy by comparing it to real-world data. For analysis, I will utilize the adjusted close stock prices of the two companies.

Definition of Variables

Open. This is the price of the stock when the trading begins for shares of firms in the systematized stock market.

High. This is the highest price of a stock during the trading day.

Low. This is the lowest price of a stock during the trading day.

Close. This represents the final price of a stock during a trading day.

Adjusted Close. This is the stock price that includes other elements like shareholders and offerings.

Data Collection

I collected data using Yahoo Finance. The data consists of 250 trading days between 19th February 2022 and 19th February 2023.

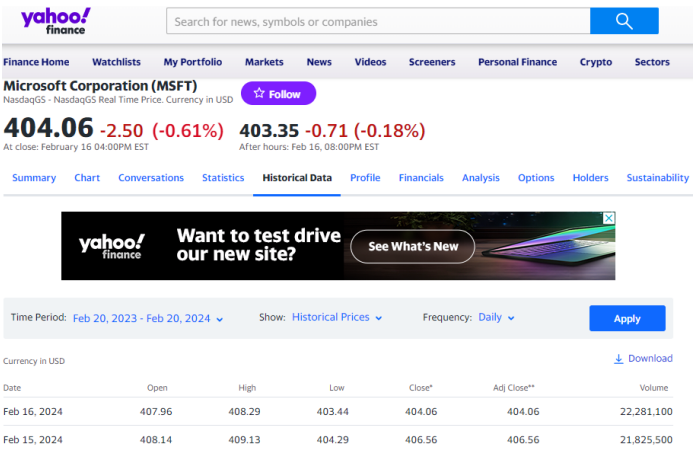


Figure 1: Microsoft Corporation

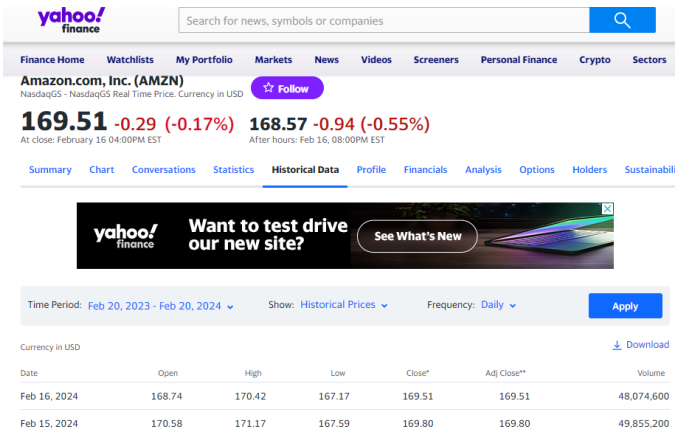


Figure 2: Amazon Inc

The collected data for the two technology companies is presented in Appendix A. I used data for 250 days which can be divided evenly in varied ways. Hence, I can choose different ranges.

Evaluation

Descriptive Statistics

I will calculate each firm's average, maximum, and minimum stock prices for the descriptive analysis.

To find the average, I will use the following formula;

$$Average = \frac{sum}{total\ number\ of\ values}$$

$$Minimum = Lowest\ stock\ price\ in\ the\ data\ set$$

$$Maximum = Highest\ stock\ price\ in\ the\ data\ set$$

Microsoft Corporation (MSFT)

	Average	Minimum	Maximum	Std Dev
Open	273.5957	246.55	306.97	16.6834
High	276.5947	250.93	308.93	16.8844
Low	271.0161	245.61	305.15	16.7394
Close	274.1341	246.27	307.26	17.1617
Adjusted Close	271.9025	244.2653	304.7588	17.022

Based on the above table, the average open price of MSFT is $273.5957, with a close price of $274.1341. The highest stock price is $276.5947, while the lowest is $271.0161. The standard deviation ranges from $16.6834 to $17.1617, indicating a highly volatile stock price change for MSFT.

Amazon Inc. (AMZN)

	Average	**Minimum**	**Maximum**	**Std Dev**
Open	99.021	89.97	108.16	4.7680
High	100.4002	92.23	110.86	4.7640
Low	97.6049	88.12	106.8	4.6030
Close	99.0571	90.73	109.82	4.6304
Adjusted Close	99.0571	90.73	109.82	4.6304

Based on the above table, the average open price of AMZN is $99.021, with a close price of $99.0571. The average highest stock price is $100.4002, while the lowest is $97.6049. The standard deviation ranges from $4.6304 to $4.7680, indicating a moderate change in AMZN stock price for the period.

Modeling a Regression Function

For this section, I will use MS Excel software to model a regression function that anticipates the stock price movements for each company. I will model the regression function using the adjusted close stock prices. I will utilize a simple linear regression line as it helps to analyze the association between two factors.[1]

The linear function is given by;

$$y = mx + c^2$$

[1] Qiao, W. (2021). Prediction model of the stock market based on Chaos Theory. *E3S Web of Conferences*, *275*, 01013. https://doi.org/10.1051/e3sconf/202127501013

[2] Almeida, A. (2021, October 26). *Examining the simple linear regression method for forecasting stock prices using Excel*. Analytics Vidhya. https://www.analyticsvidhya.com/blog/2020/10/examining-the-simple-linear-regression-method-for-forecasting-stock-prices-using-excel/

Where

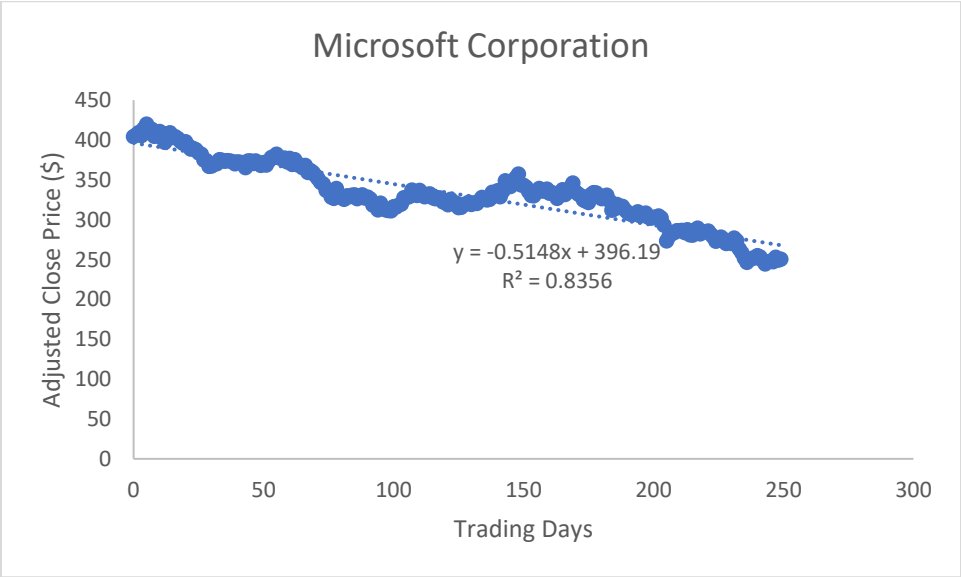
y denotes the price of the stock

m denotes the slope $m = \frac{y_2 - y_1}{x_2 - x_1}$

x denotes the time

c denotes the y-intercept

Microsoft Corporation (MSFT)



The linear function is given by;

$$y = -0.5148x + 396.19$$

According to Figure 1 above, the slope is m = -0.5148, and the y-intercept is $396.19. The above findings demonstrate that the average stock price for MSFT declines by 0.5148 dollars. It has a coefficient of determination (R^2) of 0.8356. This is a strong r-squared as it shows the accuracy level of the regression coefficient in fitting the collected data. It shows that the model can approximate 83.56% of the change in MSFT stock price.

To find the average rate of change, I will find the first differential of the linear regression. This will indicate the average stock price increase on each trading day. Hence;

$$Average\ rate\ of\ change = \frac{d}{dx}(y)$$

$$Average\ rate\ of\ change = \frac{d}{dx}(-0.5148x + 396.19)$$

$$\frac{dy}{dx} = -0.5148$$

Now, I will equate the linear regression to zero to help find the value of x;

$$-0.5148x + 396.19 = 0$$

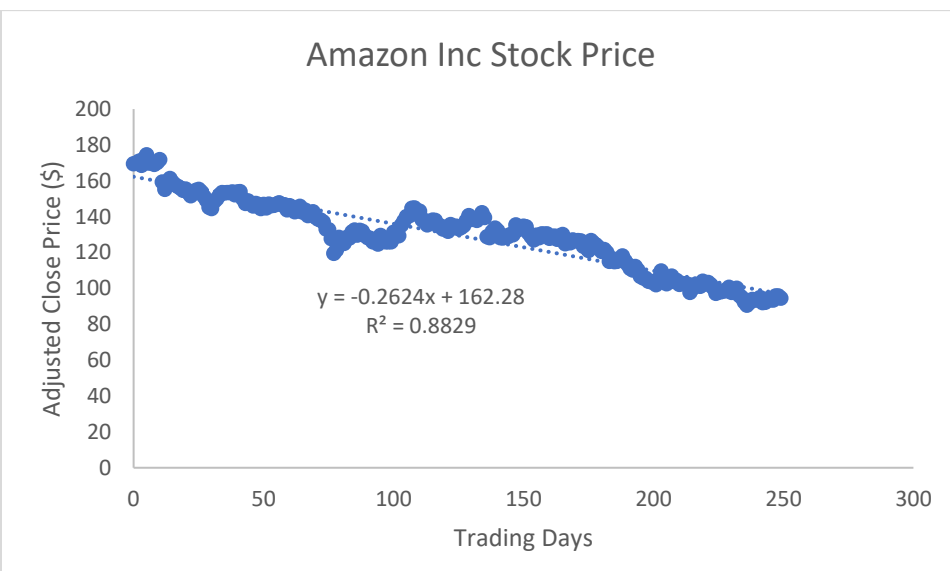
Rearranging to evaluate for x yields;

$$0.5148x = 396.19$$

$$x = 769.5998$$

Repeating the above procedure for Amazon

Amazon Inc (AMZN)



The linear function is given by;

$$y = -0.2624x + 162.28$$

According to Figure 2 above, the slope is m = -0.2624, and the y-intercept is $162.28. The above findings demonstrate that the average stock price for AMZN declines by 0.2624 dollars. It

has a coefficient of determination (R^2) of 0.8829. This is a strong r-squared as it shows the accuracy level of the regression coefficient in fitting the collected data. It shows that the model can approximate 88.29% of the change in AMZN stock price.

To find the average rate of change, I will find the first differential of the linear regression. This will indicate the average stock price increase on each trading day. Hence;

$$Average\ rate\ of\ change = \frac{d}{dx}(y)$$

$$Average\ rate\ of\ change = \frac{d}{dx}(-0.2624x + 162.28)$$

$$\frac{dy}{dx} = -0.2624$$

Now, I will equate the linear regression to zero to help find the value of x;

$$-0.2624x + 162.28 = 0$$

Rearranging to evaluate for x yields;

$$-0.2624x + 162.28 = 0$$

$$0.2624x = 162.28$$

$$x = 618.4451$$

However, the above analysis of both firms is limited. Despite the model representing the future stock prices of each firm, the predicted growth may be inaccurate. Examining the real numbers and comparing them to the x-values obtained for each function, I observed a leap in stock price decrease and increase. This shows a higher volatility than in a practical setting. Hence, the stock market is nonlinear. This nonlinear nature made me try to model it using the chaos system.

Chaos Theory

The chaotic time series forecasting is a nonlinear model that models the trajectory of a stock company.[3] It helps to create patterns from unpredictable systems. According to the chaos theory, market volatility can be attributed to environmental aspects like floods, natural disasters, and earthquakes that cause sudden drops or increases in stock prices.[4] However, stock price changes are determined by using mathematical concepts to predict price movement, changes in price volume, the complex and nonlinear personal motivation of an investor/trader, and the acceleration of changes in stock prices. Maurice Henon's map utilization provides hindsight that setting data on a 2D mapping plane as attractors subject the stock to change its price based on several factors, making the stock's data set unpredictable. Such helps us understand that equating the stock market to a practical setting analysis does not accurately predict stock price changes.

Henon aimed to reduce the 3D structure of the Lorenz attractor to a 2D. It allows us to detect the quality of stock price change in a more basic form. Essentially, the 2D structure provides a discrete system of stock market prediction entailing space folding and stretching.

The chaotic time series plot of Henon's Map has the following initial conditions and parameters;

$$x_{n+1} = 1 - ax_n^2 + y_n{}^5$$

And

$$y_{n+1} = bx_n$$

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[3] Qiao, W. (2021). Prediction model of stock market based on Chaos Theory. *E3S Web of Conferences*, *275*, 01013. https://doi.org/10.1051/e3sconf/202127501013

[4] Moses, T. (2021, July 16). *Chaos theory and market predictions*. Medium. https://medium.datadriveninvestor.com/chaos-theory-and-market-predictions-9e03ec47d67e

[5] Boeing, G. (2019, October 21). *Chaos theory and the logistic map*. Geoff Boeing. https://geoffboeing.com/2015/03/chaos-theory-logistic-map/

Where,

$a = 1.4$ and $b = 0.3$

I will use the Wolfram complexity Explorer to determine the Henon map for MSFT and AMZN. The initial condition for MSFT is (0, -0.5148), while the initial condition for AMZN is (0, -0.2624).

Henon Map for MSFT

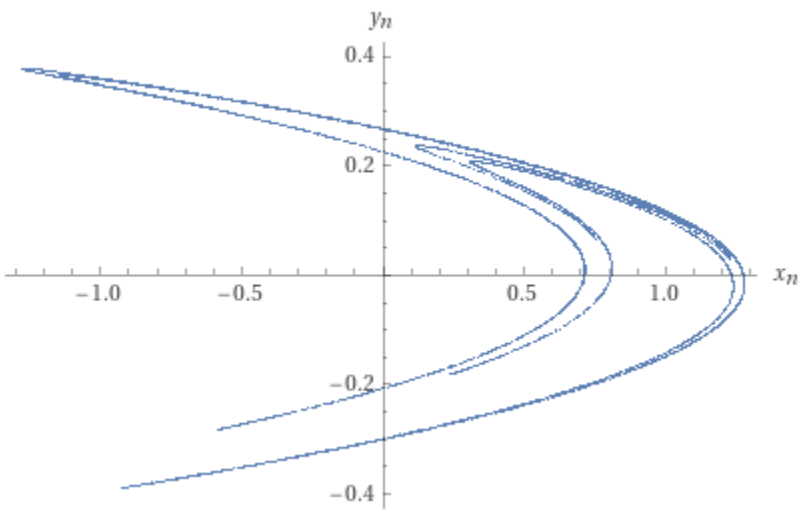

Figure 3: Wolfram Henon Map for Microsoft[6]

Henon Map for AMZN

[6] Wolfram. (2024). *Henon Map for Microsoft Corporation*. WolframAlpha computational knowledge AI. https://www.wolframalpha.com/input?i=henon%2Bmap&assumption=%7B%22C%22%2C%2B%22henon%2Bmap%22%7D%2B-%3E%2B%7B%22Formula%22%2C%2B%22dflt%22%7D

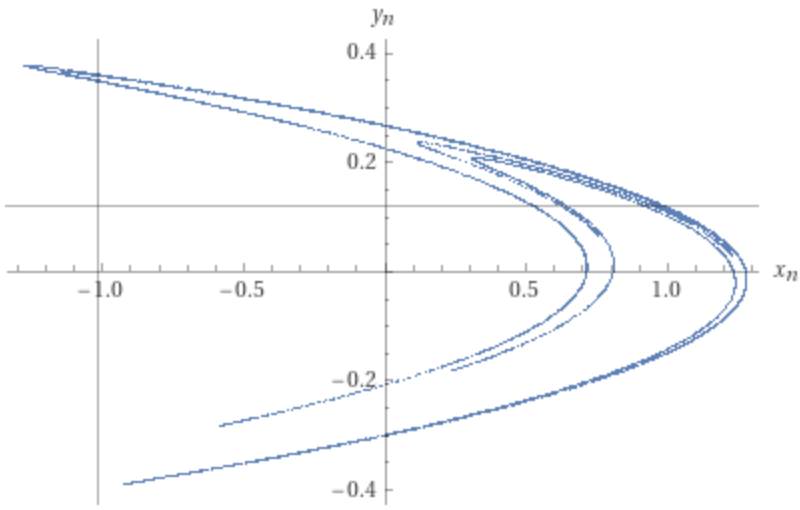

Figure 4: Wolfram Henon map for Amazon[7]

Predicting the Stock price of AMZN and MSFT Using Adjusted Close Price and Normal Distribution

In this section, I will model a normal distribution function that can be used to forecast the share price of amazon (AMZN) and Microsoft (MSFT) using adjusted close prices.

MSFT

I created a quadratic model that represents the adjusted closing price of MSFT shares. This is shown below.

[7] Wolfram Alpha. (2024). *Henon Map for Amazon Stock Price*. WolframAlpha computational knowledge AI. https://www.wolframalpha.com/input?i=henon%2Bmap&assumption=%7B%22C%22%2C%2B%22henon%2Bmap%22%7D%2B-%3E%2B%7B%22Formula%22%2C%2B%22dflt%22%7D

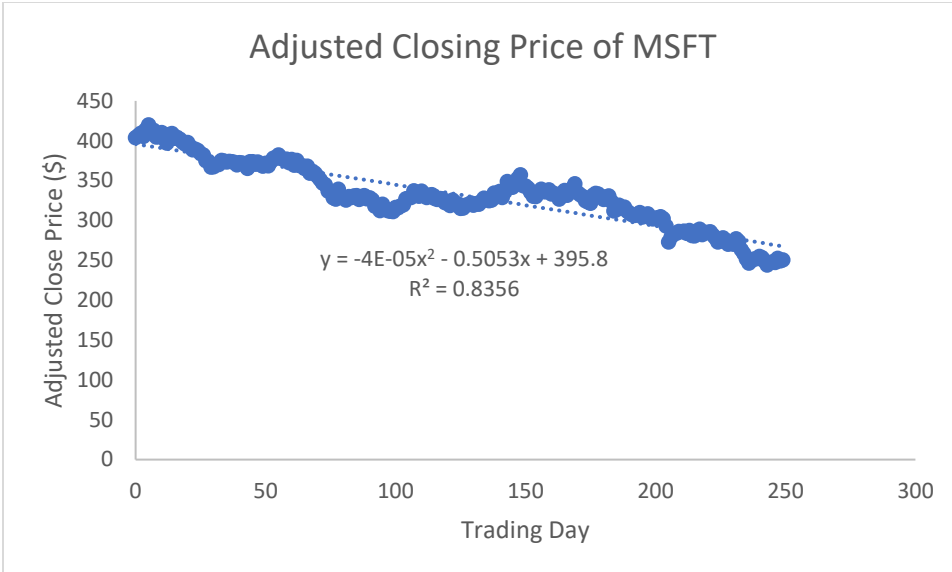

Figure 5: MSFT stock price

Figure 5 shows a downward trend in MSFT share price. From the graph, I established the following quadratic function that fits the data;

$$f(x) = -0.00004x^2 - 0.5053x + 395.8$$

To determine the average share price change, I will find the first derivative of the function as follows;

$$f(x) = -0.00004x^2 - 0.5053x + 395.8$$

$$f'(x) = \frac{d}{dx}(-0.00004x^2 - 0.5053x + 395.8)$$

$$f'(x) = -0.00008x - 0.5053$$

However, I want to determine the best trading day to sell or buy MSFT shares. Hence, I will calculate the local minima and maxima of the first derivative of the function.

First, I will equate the derivative to zero to yield;

$$-0.00008x - 0.5053 = 0$$

To find the value of x, we will use the following expression;

$$-0.00008x = 0.5053$$

Hence, x becomes;

$$x = \frac{0.5053}{-0.00008}$$

$$x = -6316.25$$

This shows that the minimum MSFT share price will be after 6,316.25 trading days before it starts to increase. Based on the function, the stock price will be $-4391.6 after these days. Since a share price cannot be a negative value, as shareholders would not pay someone to take shares off their hands, the share price would be equated to zero. This implies that at the local minimum MSFT will have no value to shareholders and would be effectively bankrupt.

I sorted the MSFT adjusted close prices in ascending order. Then, I created a graph that revealed an A-shaped distribution, demonstrating that the data is approximately normal. Therefore, I can model the MSFT share price using the normal distribution.

Hence;

$$share\ price \sim N(\mu, \sigma^2)$$

Next, I will calculate for each trading day, the daily stock return as a percentage change in the adjusted close price to yield;

$$daily\ share\ return = \frac{day\ two\ adjusted\ close\ price - day\ one\ adjusted\ close\ price}{day\ one\ adjusted\ close\ price} \times 100$$

I used Excel to compute the percentage change as illustrated in the following figure;

MSFT		
Trading Day	Adj Close	% change
0	404.06	=(J4-J3)/J3*100
1	406.56	0.7207
2	409.49	-0.957
3	405.57	2.2002
4	414.49	1.2739
5	419.77	-1.531
6	413.35	-0.014
7	413.29	-2.067
8	404.74	0.0395
9	404.9	1.3731
10	410.46	-1.809
11	403.03	-1.535
12	396.85	2.7693
13	407.84	0.2766
14	408.96	-1.413

Figure 6: Percentage change in daily stock return

Using the obtained percentage change, I used the normal distribution function to model the probabilities for the 250 days as shown below;

$$P(X = x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\times\left(\frac{X-\mu}{\sigma}\right)^2}$$

Where;

X is the random variable

x is the percentage daily return

σ is the daily return standard deviation

μ is the daily average return

To obtain the average daily return and standard deviation, I will use the percentage change data to yield;

$$\mu = -0.58076$$

$$\sigma = 6.47132$$

	% change
Mean	-0.58076
Standard Error	0.409282
Median	-0.18619
Mode	#N/A
Standard Deviation	6.471321
Sample Variance	41.878
Kurtosis	226.1543
Skewness	-14.673
Range	103.911
Minimum	-100
Maximum	3.910954
Sum	-145.19
Count	250

Figure 7: Mean and standard deviation of daily returns

Therefore, we can model the stock return to follow the normal distribution as given below;

$$Share\ return \sim N(-0.58076, 6.47132^2)$$

I will use the obtained mean and standard deviation in the normal distribution model to give;

$$P(X = x) = \frac{1}{6.47132\sqrt{2\pi}} e^{-\frac{1}{2}\times\left(\frac{X-(-0.58076)}{6.47132}\right)^2}$$

$$P(X = x) = 0.06165 \times (2.7182818285)^{-\frac{1}{2}\times\left(\frac{X-0.5069}{2.6065}\right)^2}$$

Applying the formula on the daily returns for MSFT yields;

	MSFT					
Trading Day	Adj Close	% change	Share return			Adj Close
0	404.06	0.6187	=0.06165*2.7182818285^(-0.5*((K3-$R$36)/$R$40)^2			
1	406.56	0.7207	0.060416			246.5664
2	409.49	-0.957	0.061546			247.1912
3	405.57	2.2002	0.056212			247.3896
4	414.49	1.2739	0.059169			248.1236
5	419.77	-1.531	0.060988			249.0658
6	413.35	-0.014	0.061414			249.4626
7	413.29	-2.067	0.060045			250.266
8	404.74	0.0395	0.061368			250.6131
9	404.9	1.3731	0.058903			251.6348
10	410.46	-1.809	0.060549			251.853
11	403.03	-1.535	0.060983			252.0811
12	396.85	2.7693	0.053919			252.6961
13	407.84	0.2766	0.061111			253.2118
14	408.96	-1.413	0.061142			254.779

Chart Area

Figure 8: Daily MSFT share returns

Now the daily share price return distribution becomes;

	Daily Share return
Mean	0.059882
Standard Error	0.000287
Median	0.060994
Mode	#N/A
Standard Deviation	0.004544
Sample Variance	2.06E-05
Kurtosis	123.349
Skewness	-9.87033
Range	0.06165
Minimum	3.45E-53
Maximum	0.06165
Sum	14.97059
Count	250

Figure 9: Mean and standard deviation of the daily stock return of MSFT

The normal distribution of the return is as given below;

$$P(X = x) = \frac{1}{0.0045\sqrt{2\pi}} e^{-\frac{1}{2}\times\left(\frac{X-0.05988}{0.0045}\right)^2}$$

$$P(X = x) = 88.6538 \times (2.7182818285)^{-\frac{1}{2}\times\left(\frac{X-0.05988}{0.0045}\right)^2}$$

Using the above model, I will compute the probability of the share price giving a return between 0-3%. This is a complex calculation; hence I used the Wolfram Alpha calculator to yield:

$$P = \int_0^3 \frac{1}{0.0463\sqrt{2\pi}} e^{-\frac{1}{2}\times\left(\frac{X-0.1122}{0.0463}\right)^2} = 0.90835$$

I repeated the process for AMZN.

AMZN

Creating a quadratic model that represents the adjusted closing price of AMZN shares.

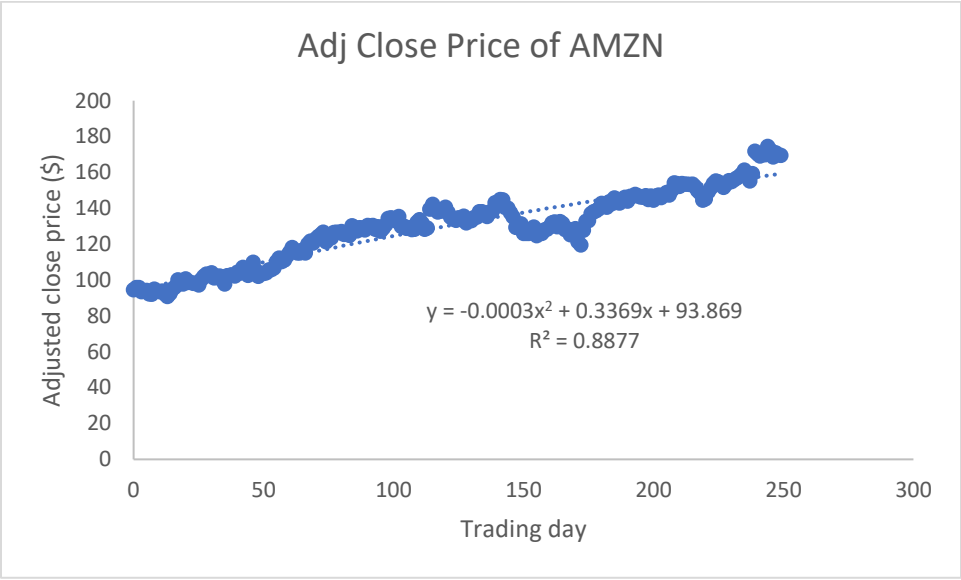

Figure 10: AMZN stock price

Figure 10 shows an upward trend in AMZN share price. From the graph, I established the following quadratic function that fits the data;

$$f(x) = -0.0003x^2 + 0.3369x + 93.869$$

To determine the average share price change, I will find the first derivative of the function as follows;

$$f(x) = -0.0003x^2 + 0.3369x + 93.869$$

$$f'(x) = \frac{d}{dx}(-0.0003x^2 + 0.3369x + 93.869)$$

$$f'(x) = -0.0006x + 0.3369$$

Similarly, I determined the best trading day to sell or buy AMZN shares. Hence, I will calculate the local minima and maxima of the first derivative of the function.

First, I will equate the derivative to zero to yield;

$$-0.0006x + 0.3369 = 0$$

To find the value of x, we will use the following expression;

$$0.0006x = 0.3369$$

Hence, x becomes;

$$x = \frac{0.3369}{0.0006}$$

$$x = 56.15$$

This shows that the maximum AMZN share price will be after 56.15 trading days before it starts to increase. Based on the function, the stock price will be $111.8401 after these days.

Similarly, I found that the data is approximately normal. Therefore, I can model the AMZN share price using the normal distribution.

Hence;

$$share\ price \sim N(\mu, \sigma^2)$$

Next, I calculated for each trading day, the daily stock return as a percentage change in the adjusted close price to yield the following figure;

AMZN			
Trading Days	Adj Close	% Change	Daily Share return
0	94.58	=(J4-J3)/J3*100	
1	95.79	0.031317	0.060198
2	95.82	-2.42121	0.056777
3	93.5	0.278077	0.060095
4	93.76	0.501281	0.059931
5	94.23	-2.18615	0.057437
6	92.17	-0.0434	0.060212
7	92.13	3.006627	0.053764
8	94.9	-1.2118	0.059449

Figure 11: Percentage change in daily stock return

Using the obtained percentage change, I used the normal distribution function to model the probabilities for the 250 days as shown below;

$$P(X = x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\times\left(\frac{X-\mu}{\sigma}\right)^2}$$

Where;

X is the random variable

x is the percentage daily return

σ is the daily return standard deviation

μ is the daily average return

To obtain the average daily return and standard deviation, I will use the percentage change data to yield;

$$\mu = -0.14815$$

$$\sigma = 6.62455$$

	% Change
Mean	-0.14815
Standard Error	0.418973
Median	0.107601
Mode	0
Standard Deviation	6.624552
Sample Variance	43.88469
Kurtosis	209.3603
Skewness	-13.8366
Range	108.2693
Minimum	-100
Maximum	8.269337
Sum	-37.0368
Count	250

Figure 12: Mean and standard deviation of daily returns

Therefore, we can model the stock return to follow the normal distribution as given below;

$$Share\ return \sim N(-0.14815, 6.62455^2)$$

I will use the obtained mean and standard deviation in the normal distribution model to give;

$$P(X = x) = \frac{1}{6.62455\sqrt{2\pi}} e^{-\frac{1}{2}\times\left(\frac{X-(-0.14815)}{6.62455}\right)^2}$$

$$P(X = x) = 0.06022 \times (2.7182818285)^{-\frac{1}{2}\times\left(\frac{X-(-0.14815)}{6.62455}\right)^2}$$

Applying the formula on the daily returns for AMZN yields:

	AMZN						Adj Close
Trading Days	Adj Close	% Change	Daily Share return		count	250	94.58
0	94.58	1.279339	=0.06022*2.7182818285^(-0.5*((K3-$S$36)/$S$40)^2)				
1	95.79	0.031317	0.060198				95.82
2	95.82	-2.42121	0.056777				93.5
3	93.5	0.278077	0.060095				93.76
4	93.76	0.501281	0.059931				94.23
5	94.23	-2.18615	0.057437				92.17
6	92.17	-0.0434	0.060212				92.13
7	92.13	3.006627	0.053764				94.9
8	94.9	-1.2118	0.059449				93.75
9	93.75	-0.21333	0.060217				93.55
10	93.55	0.395505	0.060018				93.92
11	93.92	-1.77811	0.058424				92.25
12	92.25	-1.64769	0.058697				90.73
13	90.73	1.873688	0.05748				92.43
14	92.43	2.650651	0.055078				94.88

Figure 13: Daily AMZN share returns

Now the daily share price return distribution becomes;

	Daily Share return
Mean	0.057609
Standard Error	0.000352
Median	0.05916
Mode	0.060205
Standard Deviation	0.005564
Sample Variance	3.1E-05
Kurtosis	52.71931
Skewness	-6.25881
Range	0.06022
Minimum	2.79E-51
Maximum	0.06022
Sum	14.40237
Count	250

Figure 14: Mean and standard deviation of the daily stock return of AMZN

The normal distribution of the return is as given below;

$$P(X = x) = \frac{1}{0.005564\sqrt{2\pi}} e^{-\frac{1}{2}\times\left(\frac{X-0.057609}{0.005564}\right)^2}$$

$$P(X = x) = 71.7006 \times (2.7182818285)^{-\frac{1}{2}\times\left(\frac{X-0.057609}{0.005564}\right)^2}$$

Using the above model, I will compute the probability of the share price giving a return of between 0-3%. This is a complex calculation; hence I used the Wolfram Alpha calculator to yield:

$$P = \int_0^3 \frac{1}{0.005564\sqrt{2\pi}} e^{-\frac{1}{2}\times\left(\frac{X-0.057609}{0.005564}\right)^2} = 0.849756$$

This is a good percentage return for traders, especially day traders. However, I realized that although the linear regression models the stock market, it fails to consider the volatility of the market. Hence, I decided to utilize the fractal functions to model the stock market.

Fractal analysis of the stock market

Fractal analysis is vital for investors like me to integrate in our investment decisions. The concepts allow me to identify recurring patterns across different timeframes. It helps to identify the resistance and support zones that might hold across timeframes. This will provide me with

investing confidence. To determine the fractal dimension of a fractal series, I will use the following formula;

$$D = \frac{\log(N)}{\log(S)}$$

Where,

N represents the price bars for specific price range

S denotes the scaling factor for the price range

To determine the fractal dimension, D, we will consider the stock return at time t as following a stochastic process given by $x(t)\epsilon\big(x(1), x(2), \dots x(T)\big)$. The accumulated dispersion for x(t) is given by $x(t, \tau)$ where;

$$x(t, \tau) = \sum_{t=1}^{\tau} (x(t) - (x(t))$$

The standard deviation is given by

$$S(\tau) = \sqrt{\frac{1}{\tau}\left(\sum_{t=1}^{\tau} (x(t) - (x(t))\right)^2}$$

Using the above formula and the data I collected, I created an R code that modeled the time series pattern for amazon. This is shown below.

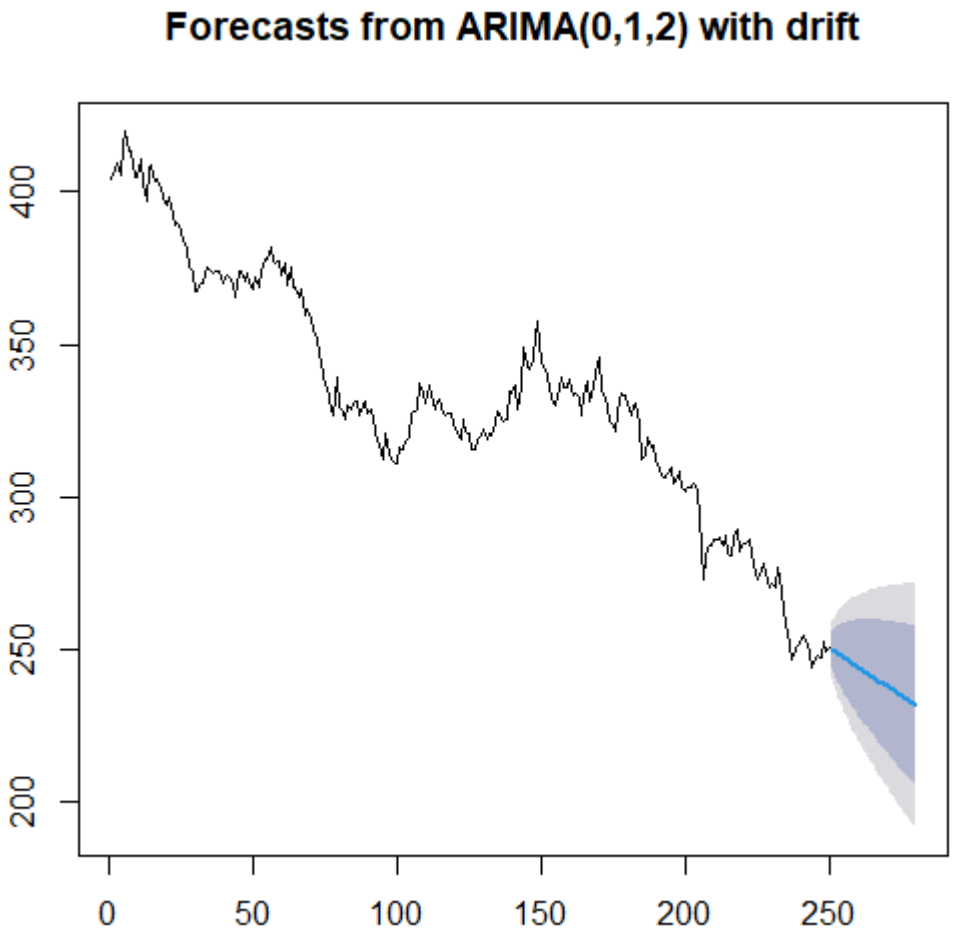

Figure 15: Fractal analysis of Amazon

Based on Figure 15, I observed that Amazon has wild swings in price actions. Hence, I decided to analyse how the stock is sensitive to slight nudges us. The greater the exponent the more sensitive the price action is for Amazon stocks.

I computed the Lyapunov exponent over 20 days periods using the below function:

$$\lambda = \frac{1}{T} \times \ln \left| \frac{S(t+T) - S(t)}{S(t)} \right| \; as \; T \to \infty$$

Where,

$S(t)$ denotes the stock price at time t.

T denotes the time horizon (20 days)

$S(t+T)$ denotes the stock price at time (t+T)

For instance, the Lyapunov exponent for the first 20 days of amazon stock price becomes;

T=0 S(0)=94.58

T=19 S(20)=97.71

$$\lambda = \frac{1}{20} \times \ln\left|\frac{97.71 - 94.58}{94.58}\right| = -0.17042$$

This is demonstrated in the following graph

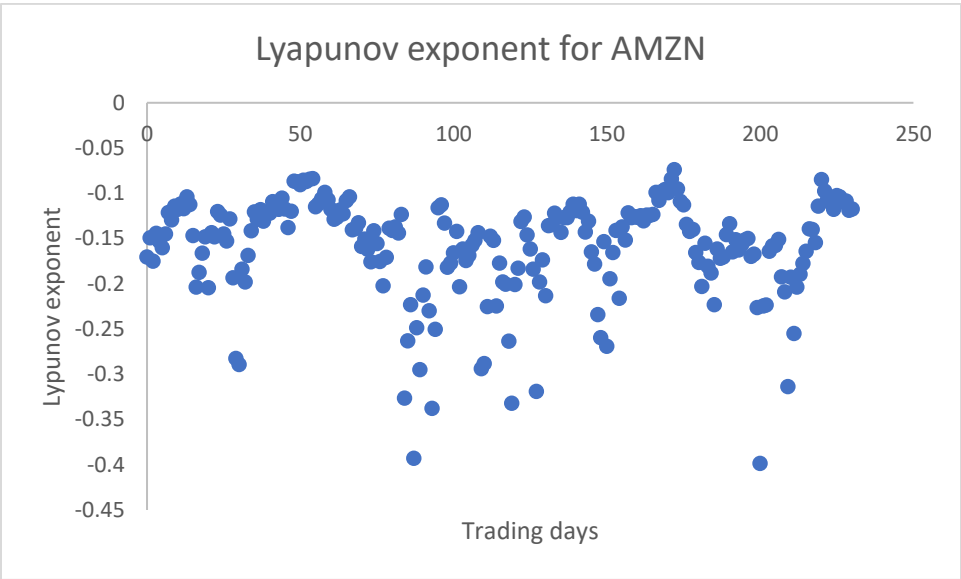

Figure 16: Lyapunov exponent for AMZN

For Amazon, all the Lyapunov exponents are negative, indicating convergence. Otherwise, positive exponents would have indicated presence of chaos and divergence. After repeating the procedure for MSFT, I noticed that it also had negative exponents.

A Monte Carlo Simulation of Stock Prices

Monte Carlo simulation is a powerful statistical method that investors use to predict the stock movement and make informed investment decision in the market. The technique helps to predict stock prices using historical data. Therefore, I decided to utilize it to help in making informed investment decisions regarding MSFT and AMZN. I chose this approach because it is a

computational technique that utilizes historical data and random sampling to model stock prices. This helps to determine the probability of different scenarios in the stock market. Since this is a computational program, I utilized R-program to simulate the stock prices. I leveraged the power of R-program and its library ecosystem.

I understand that Monte Carlo simulation follows a random walk. Hence it is technically a Markov process where although past stock price information is incorporated, it does not inform future price movements because they are conditionally independent.

I will use the following formula to run the simulation;

$$\frac{\Delta s}{s} = \mu \Delta t + \sigma \epsilon \sqrt{\Delta t}$$[8]

Where,

Δt denotes the elapsed time

σ is the standard deviation of returns

Δs denotes the change in stock price

s denotes the stock price

ϵ denotes the random variable

μ is the expected return

To obtain the change in stock price after a given period, I will make Δs the subject of the expression to yield;

$$\Delta s = S \times \left(\mu \Delta t + \sigma \epsilon \sqrt{\Delta t} \right)$$

Thus, to determine today's price, we will utilize the following expression;

[8] Harper, D. R. (2022). *How to use Monte Carlo simulation with GBM*. Investopedia. https://www.investopedia.com/articles/07/montecarlo.asp

$$Price\ today\ (S_t) = Yesterday\ Price(S_{t-1}) \times e^r$$

But we need the daily return

Amazon Inc

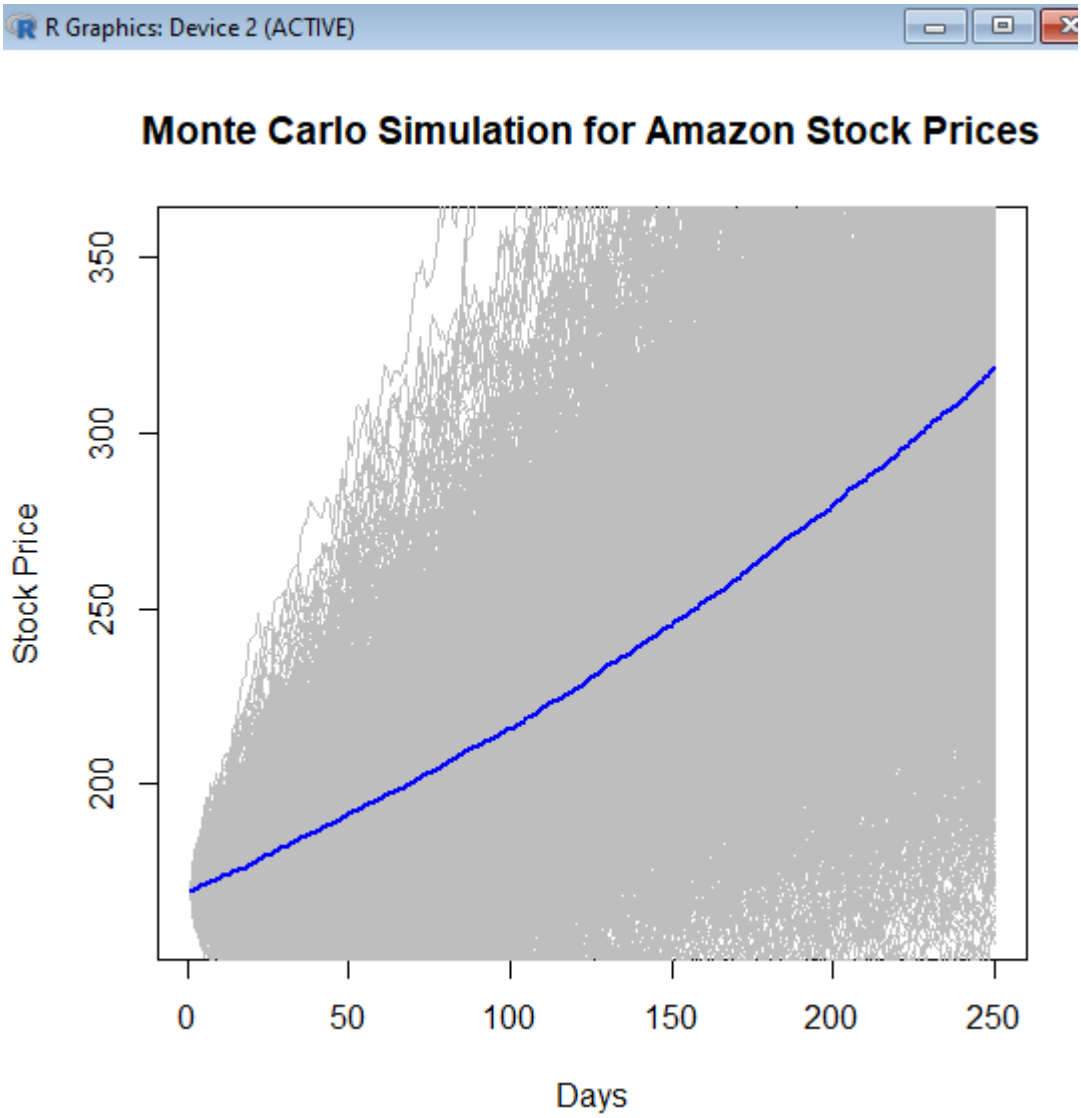

Figure 17: A Monte Carlo simulation of Amazon Stock Price

Figure 15 above displays the Monte Carlo simulation of AMZN stock prices. The program I created for the simulation is in Appendix C.

Microsoft Corporation

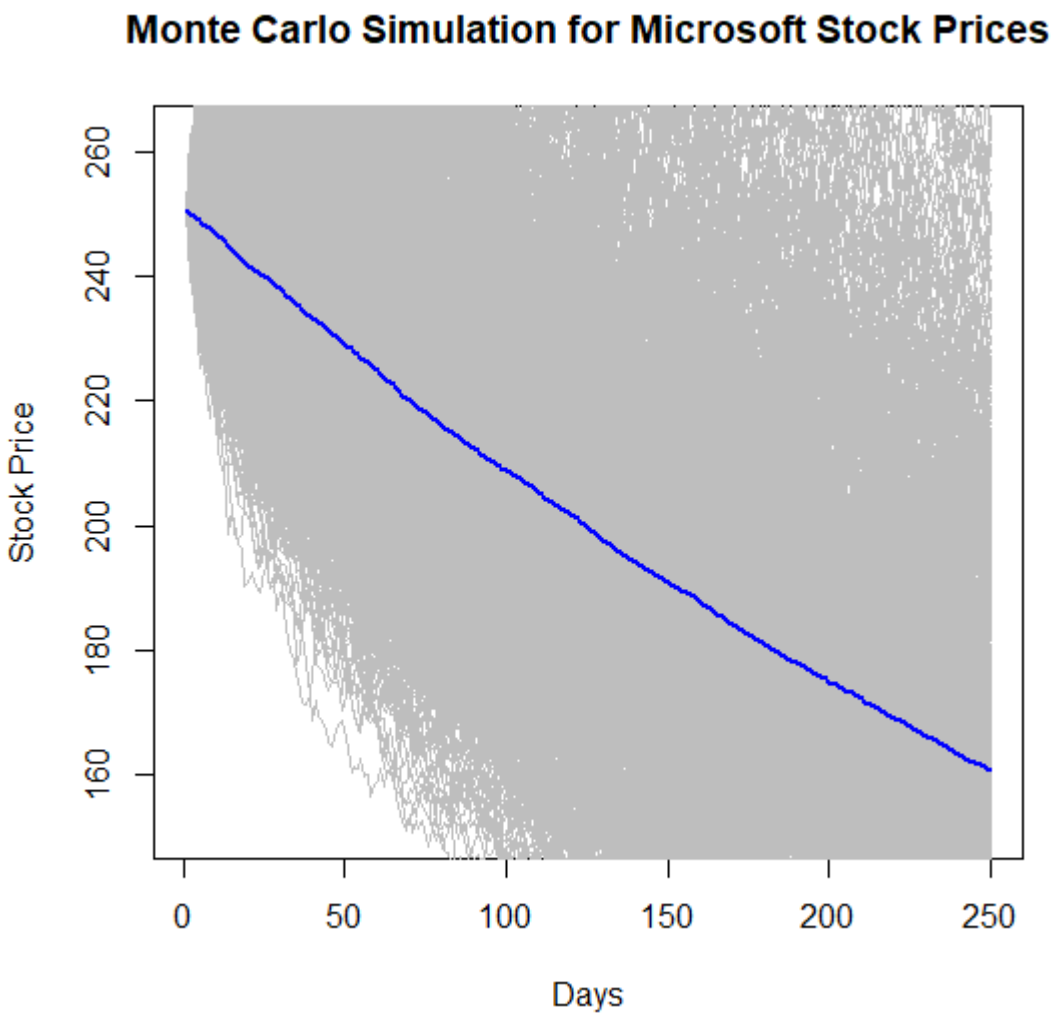

Figure 18: A Monte Carlo simulation of MSFT stock prices

Figure 16 above displays the Monte Carlo simulation of MSFT stock prices. The program I created for the simulation is in Appendix D.

Conclusion

This exploration has analyzed the stock market using the chaos model. The analyses provided relevant coefficients of determination, revealing the suitability of the linear regression model in fitting the data set. Observing how the stock models to forecast the stock prices are developed using Henon's Map and the regression model makes it challenging to predict the prices

accurately. This is attributed to external factors that contribute to the change in stock price, such as social, political, and environmental factors. It makes forecasting stock price, not a definitive process. Hence, the stock price becomes chaotic, as observed using Henon's map. The normal distribution model provided detailed insights into the price stock return of 84.98% for AMZN and 90.84%. Nevertheless, the exploration has provided insights regarding the unpredictability of the stock market movements due to certain situations (external) influencing the market.

Evaluation and Limitation

A limitation of the investigation is that I used a small historical data sample for analysis. This affects the objectivity strength of the model. In addition, the linear regression model fails to consider the randomness of the stock market system. This makes it challenging to forecast long-term data. Future research should consider using large sample data for stock companies to provide an accurate model predicting prices. Another limitation is that fractal analysis cannot be used independently to make investment decisions. This is due to the different generated patterns making it difficult to determine the exit and entry points. Hence, they need to be enhanced by combining with other technical analysis tools.

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1%2F2%C3%97%28%28X-
0.057609%29%2F0.05564%29%5E2%2B%29%2B%E3%80%97

Appendix A: Data set

Amazon Inc

Date	Open	High	Low	Close	Adj Close	Volume
2/21/2023	95.34	95.61	94.27	94.58	94.58	56580400
2/22/2023	95.1	97.01	94.8	95.79	95.79	59534100
2/23/2023	96.12	96.43	93.67	95.82	95.82	48467000
2/24/2023	93.53	94.14	92.32	93.5	93.5	57053800
2/27/2023	94.28	94.78	93.14	93.76	93.76	47470300
2/28/2023	93.14	94.69	92.92	94.23	94.23	43959300
3/1/2023	93.87	94.68	91.59	92.17	92.17	52299500
3/2/2023	91.41	92.23	90.39	92.13	92.13	55509400
3/3/2023	92.74	94.94	92.66	94.9	94.9	55759600
3/6/2023	95.19	96.55	93.74	93.75	93.75	52112400
3/7/2023	94.06	95.09	92.78	93.55	93.55	49100700
3/8/2023	93.6	94.17	92.18	93.92	93.92	44899100
3/9/2023	93.68	96.21	92.18	92.25	92.25	56218700
3/10/2023	92.67	93.57	90.25	90.73	90.73	69827500
3/13/2023	89.97	94.02	88.12	92.43	92.43	72397100
3/14/2023	93.83	95.07	92.71	94.88	94.88	60912700
3/15/2023	93.22	96.67	93.07	96.2	96.2	70731800
3/16/2023	95.75	100.99	95.61	100.04	100.04	84446900
3/17/2023	99.79	100.66	97.46	98.95	98.95	87300200
3/20/2023	98.41	98.48	95.7	97.71	97.71	62388900
3/21/2023	98.14	100.85	98	100.61	100.61	58597300
3/22/2023	100.45	102.1	98.61	98.7	98.7	57475400
3/23/2023	100.43	101.06	97.62	98.71	98.71	57559300
3/24/2023	98.07	98.3	96.4	98.13	98.13	56095400
3/27/2023	99.07	99.34	97.08	98.04	98.04	46721300
3/28/2023	98.11	98.44	96.29	97.24	97.24	38720100
3/29/2023	98.69	100.42	98.56	100.25	100.25	49783300
3/30/2023	101.55	103.04	101.01	102	102	53633400
3/31/2023	102.16	103.49	101.95	103.29	103.29	56750300
4/3/2023	102.3	103.29	101.43	102.41	102.41	41135700
4/4/2023	102.75	104.2	102.11	103.95	103.95	48662500
4/5/2023	103.91	103.91	100.75	101.1	101.1	45175400
4/6/2023	100.75	102.38	99.8	102.06	102.06	43808000
4/10/2023	100.96	102.2	99.57	102.17	102.17	37261200
4/11/2023	100.8	101	99.01	99.92	99.92	60417800
4/12/2023	100.4	100.51	97.71	97.83	97.83	56735000
4/13/2023	98.95	102.57	98.71	102.4	102.4	67925100
4/14/2023	102.07	103.2	101.11	102.51	102.51	51450500
4/17/2023	103.16	103.73	101.59	102.74	102.74	39919500
4/18/2023	103.95	104.2	101.52	102.3	102.3	39790500
4/19/2023	101.58	105.12	101.39	104.3	104.3	58398900
4/20/2023	103.53	105.25	103.21	103.81	103.81	57696900
4/21/2023	106.1	108.15	105.08	106.96	106.96	86774200
4/24/2023	107.66	109.23	105.07	106.21	106.21	69575600
4/25/2023	104.91	105.45	102.45	102.57	102.57	65026800
4/26/2023	105.04	106.62	104.1	104.98	104.98	73803800
4/27/2023	108.16	110.86	106.8	109.82	109.82	149961200
4/28/2023	107.73	109.48	104.33	105.45	105.45	130565000
5/1/2023	104.95	105.23	101.82	102.05	102.05	74728100
5/2/2023	101.47	103.9	101.15	103.63	103.63	73469400
5/3/2023	103.74	105.96	103.28	103.65	103.65	65051900
5/4/2023	104.04	105.39	103.31	104	104	45345500
5/5/2023	104.27	105.76	103.55	105.66	105.66	56912900
5/8/2023	105.04	106.1	104.7	105.83	105.83	49430900
5/9/2023	105.48	106.79	105.16	106.62	106.62	44089400
5/10/2023	108.1	110.67	108.05	110.19	110.19	78627600
5/11/2023	111.03	113.28	110.49	112.18	112.18	74924800
5/12/2023	112.16	112.64	109.32	110.26	110.26	49810100
5/15/2023	111.15	112.29	109.25	111.2	111.2	53011100
5/16/2023	111.05	114.79	111.05	113.4	113.4	71472900
5/17/2023	114.89	115.83	114.22	115.5	115.5	65655200
5/18/2023	116.69	118.6	116.34	118.15	118.15	73174100
5/19/2023	118.16	118.31	115.7	116.25	116.25	54990200
5/22/2023	116.77	116.77	114.25	115.01	115.01	70741100
5/23/2023	114.27	117.14	113.78	114.99	114.99	67576300
5/24/2023	115.35	117.34	115.02	116.75	116.75	63487900
5/25/2023	116.63	116.87	114.31	115	115	66496700
5/26/2023	116.04	121.5	116.02	120.11	120.11	96779900
5/30/2023	122.37	122.92	119.86	121.66	121.66	64314800
5/31/2023	121.45	122.04	119.17	120.58	120.58	72800800
6/1/2023	120.69	123.49	119.93	122.77	122.77	54375100
6/2/2023	124.92	126.39	124.02	124.25	124.25	61215000
6/5/2023	123.36	125.8	123.03	125.3	125.3	47950100
6/6/2023	125.07	127.4	125	126.61	126.61	45695200
6/7/2023	127.01	127.37	120.63	121.23	121.23	95663300
6/8/2023	123.01	125.63	122.26	124.25	124.25	62159300

Microsoft Corporation

Date	Open	High	Low	Close	Adj Close
2/21/2023	254.48	255.49	251.59	252.67	250.6131
2/22/2023	254.09	254.34	250.34	251.51	249.4626
2/23/2023	255.56	256.84	250.48	254.77	252.6961
2/24/2023	249.96	251	248.1	249.22	247.1912
2/27/2023	252.46	252.82	249.39	250.16	248.1236
2/28/2023	249.07	251.49	248.73	249.42	247.3896
3/1/2023	250.76	250.93	245.79	246.27	244.2653
3/2/2023	246.55	251.4	245.61	251.11	249.0658
3/3/2023	252.19	255.62	251.39	255.29	253.2118
3/6/2023	256.43	260.12	255.98	256.87	254.779
3/7/2023	256.3	257.69	253.39	254.15	252.0811
3/8/2023	254.04	254.54	250.81	253.7	251.6348
3/9/2023	255.82	259.56	251.58	252.32	250.266
3/10/2023	251.08	252.79	247.6	248.59	246.5664
3/13/2023	247.4	257.91	245.73	253.92	251.853
3/14/2023	256.75	261.07	255.86	260.79	258.6671
3/15/2023	259.98	266.48	259.21	265.44	263.2792
3/16/2023	265.21	276.56	263.28	276.2	273.9516
3/17/2023	278.26	283.33	276.32	279.43	277.1553
3/20/2023	276.98	277.48	269.85	272.23	270.0139
3/21/2023	274.88	275	269.52	273.78	271.5513
3/22/2023	273.4	281.04	272.18	272.29	270.0735
3/23/2023	277.94	281.06	275.2	277.66	275.3997
3/24/2023	277.24	280.63	275.28	280.57	278.286
3/27/2023	280.5	281.46	275.52	276.38	274.1301
3/28/2023	275.79	276.14	272.05	275.23	272.9895
3/29/2023	278.96	281.14	278.41	280.51	278.2265
3/30/2023	284.23	284.46	281.48	284.05	281.7377
3/31/2023	283.73	289.27	283	288.3	285.9531
4/3/2023	286.52	288.27	283.95	287.23	284.8918
4/4/2023	287.23	290.45	285.67	287.18	284.8422
4/5/2023	285.85	287.15	282.92	284.34	282.0253
4/6/2023	283.21	292.08	282.03	291.6	289.2262
4/10/2023	289.21	289.6	284.71	289.39	287.0342
4/11/2023	285.75	285.98	281.64	282.83	280.5276
4/12/2023	284.79	287.01	281.96	283.49	281.1822
4/13/2023	283.59	289.9	283.17	289.84	287.4806
4/14/2023	287	288.48	283.69	286.14	283.8107
4/17/2023	289.93	291.6	286.16	288.8	286.449
4/18/2023	291.57	291.76	287.01	288.37	286.0225
4/19/2023	285.99	289.05	284.54	288.45	286.1019
4/20/2023	285.25	289.03	285.08	286.11	283.7809
4/21/2023	285.01	286.27	283.06	285.76	283.4338
4/24/2023	282.09	284.95	278.72	281.77	279.4763
4/25/2023	279.51	281.6	275.37	275.42	273.1779
4/26/2023	296.7	299.57	292.73	295.37	292.9655
4/27/2023	295.97	305.2	295.25	304.83	302.3485
4/28/2023	304.01	308.93	303.31	307.26	304.7588
5/1/2023	306.97	308.6	305.15	305.56	303.0726
5/2/2023	307.76	309.18	303.91	305.41	302.9238
5/3/2023	306.62	308.61	304.09	304.4	301.9221
5/4/2023	306.24	307.76	303.4	305.41	302.9238
5/5/2023	305.72	311.97	304.27	310.65	308.1212
5/8/2023	310.13	310.2	306.09	308.65	306.1375
5/9/2023	308	310.04	306.31	307	304.5009
5/10/2023	308.62	313	307.67	312.31	309.7676
5/11/2023	310.1	311.12	306.26	310.11	307.5855
5/12/2023	310.55	310.65	306.6	308.97	306.4548
5/15/2023	309.1	309.9	307.59	309.46	306.9409
5/16/2023	309.83	313.71	309.83	311.74	309.2023
5/17/2023	312.29	314.43	310.74	314	312.1247
5/18/2023	314.53	319.04	313.72	318.52	316.6177
5/19/2023	316.74	318.75	316.37	318.34	316.4388
5/22/2023	318.6	322.59	318.01	321.18	319.2618
5/23/2023	320.03	322.72	315.25	315.26	313.3772
5/24/2023	314.73	316.5	312.61	313.85	311.9756
5/25/2023	323.24	326.9	320	325.92	323.9735
5/26/2023	324.02	333.4	323.88	332.89	330.9019
5/30/2023	335.23	335.74	330.52	331.21	329.232
5/31/2023	332.29	335.94	327.33	328.39	326.4288
6/1/2023	325.93	333.53	324.72	332.58	330.5938
6/2/2023	334.25	337.5	332.55	335.4	333.3969
6/5/2023	335.22	338.56	334.66	335.94	333.9337
6/6/2023	335.33	335.37	332.17	333.68	331.6872
6/7/2023	331.65	334.49	322.5	323.38	321.4487
6/8/2023	323.94	326.64	323.35	325.26	323.3175
6/9/2023	324.99	329.99	324.41	326.79	324.8383
6/12/2023	328.58	332.1	325.16	331.85	329.8681
6/13/2023	334.47	336.98	330.39	334.29	332.2935

Appendix B: R Codes for MSFT and AMZN

AMZN

```
# Number of simulations and days
num_simulations <- 1000
num_days <- 250
# Run Monte Carlo simulation
simulations <- monte_carlo_simulation(tesla_data, num_simulations, num_days)
# Plot the simulations
plot(1:num_days, simulations[, 1], type = "l", col = "gray", xlab = "Days", ylab = "Stock Price",
     main = "Monte Carlo Simulation for Tesla Stock Prices")
for (i in 2:num_simulations) {
  lines(1:num_days, simulations[, i], col = "gray")
}
lines(1:num_days, apply(simulations, 1, mean), col = "blue", lwd = 2)
# Create Amazon data
amazon_data <- data.frame(
  Trading_Days = 0:249,
  Adj_Close = c(
    94.580002, 95.790001, 95.82, 93.5, 93.760002, 94.230003, 92.169998, 92.129997, 94.900002, 93.75,
    93.550003, 93.919998, 92.25, 90.730003, 92.43, 94.879997, 96.199997, 100.040001, 98.949997, 97.709999,
    100.610001, 98.699997, 98.709999, 98.129997, 98.040001, 97.239998, 100.25, 102, 103.290001, 102.410004,
    103.949997, 101.099998, 102.059998, 102.169998, 99.919998, 97.830002, 102.400002, 102.510002, 102.739998,
    102.300003, 104.300003, 103.809998, 106.959999, 106.209999, 102.57, 104.980003, 109.82, 105.449997, 102.050003,
    103.629997, 103.650002, 104, 105.660004, 105.830002, 106.620003, 110.190002, 112.18, 110.260002, 111.199997,
    113.400002, 115.5, 118.150002, 116.25, 115.010002, 114.989998, 116.75, 115, 120.110001, 121.660004, 120.580002,
    122.769997, 124.25, 125.300003, 126.610001, 121.230003, 124.25, 123.43, 126.57, 126.660004, 126.419998, 127.110001,
    125.489998, 125.779999, 124.830002, 130.149994, 129.330002, 127.330002, 129.179993, 129.039993, 127.900002,
    130.360001, 130.220001, 130.380005, 128.360001, 129.779999, 127.129997, 128.779999, 130.800003, 134.300003,
    134.679993, 133.559998, 132.830002, 135.360001, 129.960007, 130, 128.800003, 129.130005, 128.149994, 128.25,
    132.210007, 133.679993, 131.690002, 128.210007, 128.910004, 139.570007, 142.220001, 139.940002, 137.850006,
    138.559998, 138.410004, 140.570007, 137.669998, 135.070007, 133.979996, 133.220001, 134.679993, 134.25, 135.520004,
    131.839996, 133.259995, 133.139999, 134.910004, 135.070007, 138.009995, 138.119995, 137.270004, 135.360001,
    137.850006, 138.229996, 143.100006, 141.229996, 144.850006, 144.720001, 140.389999, 139.979996, 137.630005,
    135.289993, 129.330002, 129.119995, 131.270004, 125.980003, 125.980003, 125.980003, 127.120003, 129.460007,
    124.720001, 127, 125.959999, 127.959999, 128.259995, 129.479996, 131.830002, 132.330002, 129.789993, 132.550003,
    131.470001, 128.130005, 128.399994, 125.169998, 126.559998, 128.559998, 121.389999, 119.57, 127.739998, 132.710007,
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    145.800003, 143.199997, 142.830002, 145.179993, 146.130005, 143.899994, 146.710007, 146.740005, 147.729996,
```


```
    130.360001, 130.220001, 130.380005, 128.360001, 129.779999, 127.129997, 128.779999, 130.800003, 134.300003,
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    133.089996, 137, 138.070007, 138.600006, 139.740005, 142.710007, 142.080002, 140.600006, 143.559998, 142.589996,
    145.800003, 143.199997, 142.830002, 145.179993, 146.130005, 143.899994, 146.710007, 146.740005, 147.729996,
```


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    147.029999, 146.320007, 146.089996, 147.029999, 144.839996, 146.880005, 144.520004, 146.880005, 147.419998,
    145.889999, 147.479996, 148.839996, 147.419998, 149.970001, 154.070007, 153.789993, 152.119995, 153.839996,
    153.419998, 153.410004, 153.339996, 153.380005, 151.940002, 149.929993, 148.470001, 144.570007, 145.240005,
    149.100006, 151.369995, 153.729996, 155.179993, 154.619995, 153.160004, 151.710007, 153.5, 155.339996, 154.779999,
    156.020004, 156.869995, 157.75, 159.119995, 161.259995, 159, 155.199997, 159.279999, 171.809998, 170.309998,
    169.149994, 170.529999, 169.839996, 174.449997, 172.339996, 168.639999, 170.979996, 169.800003, 169.509995
  )
)
# Monte Carlo simulation function
monte_carlo_simulation <- function(data, num_simulations, num_days) {
  simulations <- matrix(0, nrow = num_days, ncol = num_simulations)
  simulations[1, ] <- data$Adj_Close[length(data$Adj_Close)]
  for (i in 2:num_days) {
    daily_returns <- diff(log(data$Adj_Close))
    mean_return <- mean(daily_returns)
    std_dev_return <- sd(daily_returns)
    simulations[i, ] <- simulations[i - 1, ] * exp(mean_return + std_dev_return * rnorm(num_simulations))
  }
  return(simulations)
}
```

MSFT

```
# Number of simulations and days
num_simulations <- 1000
num_days <- 250
# Run Monte Carlo simulation
simulations <- monte_carlo_simulation(amazon_data, num_simulations, num_days)
# Plot the simulations
plot(1:num_days, simulations[, 1], type = "l", col = "gray", xlab = "Days", ylab = "Stock Price",
     main = "Monte Carlo Simulation for Amazon Stock Prices")
for (i in 2:num_simulations) {
  lines(1:num_days, simulations[, i], col = "gray")
}
lines(1:num_days, apply(simulations, 1, mean), col = "blue", lwd = 2)
# Create Microsoft data
msft_data <- data.frame(
  Trading_Day = 0:249,
  Adj_Close = c(
    404.059998, 406.559998, 409.48999, 405.570007, 414.4935, 419.773712, 413.345612, 413.285706, 404.741516, 404.901245,
    410.460968, 403.034698, 396.84613, 407.835815, 408.963715, 403.184418, 404.122681, 401.816925, 398.163696, 395.778107,
    397.934143, 393.142975, 388.751099, 389.549622, 387.75296, 383.920044, 382.063446, 375.096375, 373.998383, 367.071198,
    367.260834, 369.915924, 370.185425, 375.345886, 374.58728, 373.379547, 373.968445, 373.88858, 372.850525, 369.935883,
    372.571045, 371.962128, 370.045715, 365.254547, 373.678955, 373.688965, 370.614624, 373.539246, 370.265289, 368.119232,
    371.832367, 368.458649, 373.818726, 378.210602, 378.150696, 381.993622, 377.911133, 376.733307, 377.152557, 372.381378,
    376.743317, 369.167328, 375.475677, 368.987671, 368.837921, 365.26181, 368.240265, 359.294983, 361.795288, 359.13562,
    355.151062, 351.435486, 346.972839, 344.731537, 336.802307, 336.005402, 328.534424, 326.621857, 339.352448, 329.251648,
    328.046326, 325.406586, 330.038574, 328.833252, 330.775696, 331.353485, 326.462494, 329.879211, 331.134338, 327.119934,
    328.544373, 325.994293, 318.124817, 317.726379, 312.177917, 320.555389, 314.528809, 312.426971, 311.580231, 310.93277,
    316.31189, 315.783936, 318.294189, 319.529388, 327.378876, 327.787292, 328.94281, 337.390015, 334.760254, 330.486816,
    336.632965, 332.977142, 328.634033, 331.59256, 332.259918, 327.388855, 326.492371, 327.518372, 327.139832, 322.448059,
    321.730835, 318.732483, 325.735291, 321.21283, 320.635101, 315.255981, 315.654449, 319.160797, 319.937775, 322.104767,
    319.092865, 321.001404, 320.305603, 324.102753, 328.138519, 325.822418, 324.709137, 325.544098, 334.331299, 333.913818,
    336.349182, 328.744843, 335.752747, 348.883881, 343.04892, 341.716919, 344.798431, 352.959351, 357.343048, 343.665222,
    343.178131, 340.613586, 335.186188, 330.484436, 329.848206, 335.206055, 339.231873, 336.130493, 335.971436, 338.506226,
    333.048981, 333.844238, 332.571869, 326.637543, 333.019165, 337.681152, 331.567902, 336.031097, 340.285522, 346.021088,
    335.325348, 332.293549, 329.868103, 324.838348, 323.317505, 321.4487, 331.687195, 333.933716, 333.396881, 330.59375,

    326.428802, 329.231964, 330.901947, 323.973541, 311.975616, 313.377228, 319.261841, 316.438812, 316.617706, 312.124725,
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    302.923798, 303.072601, 304.758789, 302.348541, 292.965515, 273.177948, 279.476257, 283.433807, 283.780914, 286.101868,
    286.022522, 286.449005, 283.810699, 287.48056, 281.18222, 280.527618, 287.034241, 289.226227, 282.02533, 284.842224,
    284.891815, 285.953064, 281.737701, 278.226532, 272.989532, 274.130127, 278.286041, 275.399719, 270.073456, 271.5513,
    270.013947, 277.155273, 273.95163, 263.279175, 258.667053, 251.852966, 246.56636, 250.266006, 251.634766, 252.081085,
    254.778961, 253.211807, 249.065842, 244.265259, 247.389603, 248.123581, 247.191223, 252.69606, 249.462585, 250.613144
  )
)
# Monte Carlo simulation function
monte_carlo_simulation <- function(data, num_simulations, num_days) {
  simulations <- matrix(0, nrow = num_days, ncol = num_simulations)
  simulations[1, ] <- data$Adj_Close[length(data$Adj_Close)]
  for (i in 2:num_days) {
    daily_returns <- diff(log(data$Adj_Close))
    mean_return <- mean(daily_returns)
    std_dev_return <- sd(daily_returns)
    simulations[i, ] <- simulations[i - 1, ] * exp(mean_return + std_dev_return * rnorm(num_simulations))
  }
  return(simulations)
}
```